

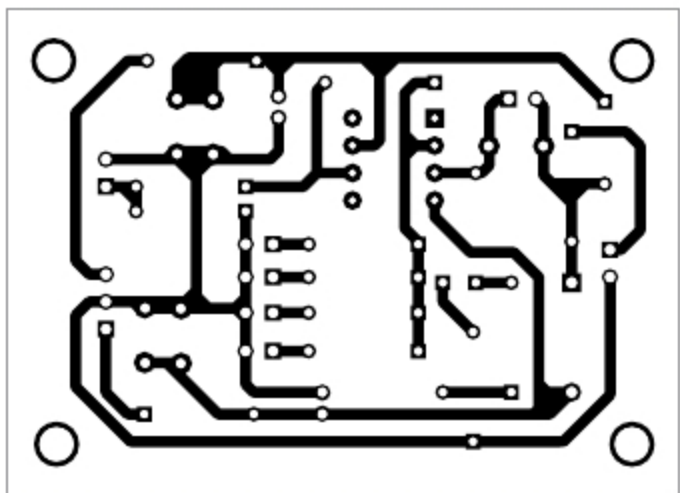


Fig. 2: Actual-size PCB layout of the active probe

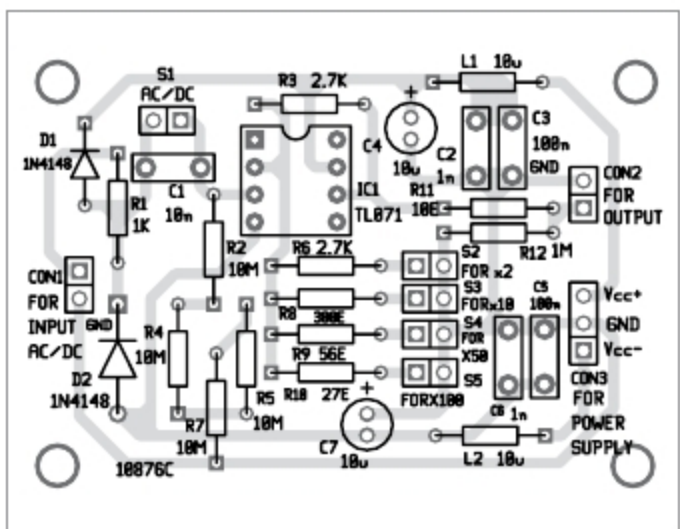


Fig. 3: Components layout for the PCB

requires two power supplies: positive terminal at pin 7 and negative terminal at pin 4 of IC1. Depending on the op-amp used, you may use power supplies as 2x 3V, 2x 3.3V, 2x 4.5V, 2x 5V, 2x 9V (two 6F22 batteries), etc.

Also, depending on the op-amp, the active probe can have low power supply current (for example, MCP601 or similar op-amp), so the battery will work hundreds of hours without requiring to be replaced. Depending on the op-amp and the required amplitude of the output signal, power supplies can be from $\pm 3V$ for MCP601, to $\pm 18V$ for OPA132 and OPA134.

Power supplies are filtered with LC filters to reduce the noise introduced by external power supplies working from AC mains or from DC/DC converters.

Construction and testing

An actual-size PCB layout of the active probe is shown in Fig. 2

and its components layout in Fig. 3. Assemble the circuit on the PCB. Connect AC/DC input across CON1, so output is available at CON2.

Connect the dual-power supply depending on your requirement. Alternatively, you can use two 9V batteries (6F22) as the dual-power supply.

The circuit can be used as a universal high-impedance buffer with selectable gain. Care should be taken while using the probe because the high input resistance of the probe can capture a lot of electromagnetic noise. **EFY**

EFY note. The circuit and PCB are designed for op-amp TL071 or equivalent op-amp. If another op-amp is used, modify it accordingly.



Petre Tzv Petrov was a researcher and assistant professor in Technical University of Sofia, Bulgaria and expert-lecturer in OFPPT (Casablanca), Kingdom of Morocco. Now, he is working as an electronics engineer in the private sector in Bulgaria

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